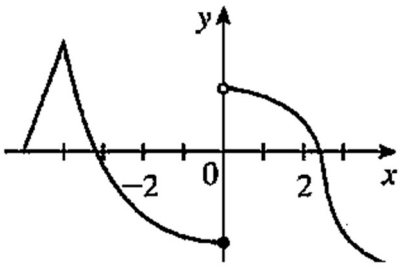
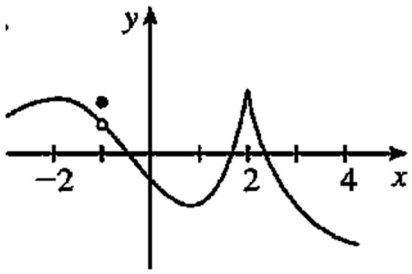
Worksheet 2-4 A Differentiability and Continuity #1
AP Calculus Name_____

For questions 1– 4, the graph of *f* is given. State, with reasons, the numbers at which *f* is not differentiable.

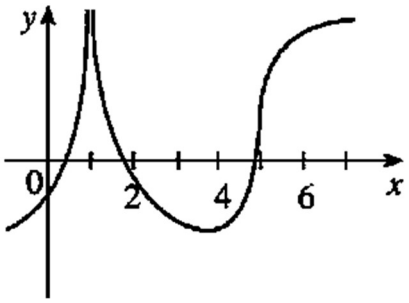
1.



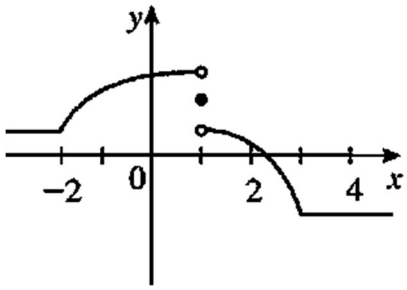
2.



3.



4.



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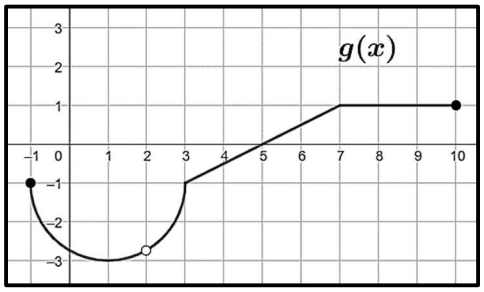
5. The graph of $g(x)$ is given at the right. Find all x values in the open interval $(-1, 10)$ where $g(x)$ is not differentiable.



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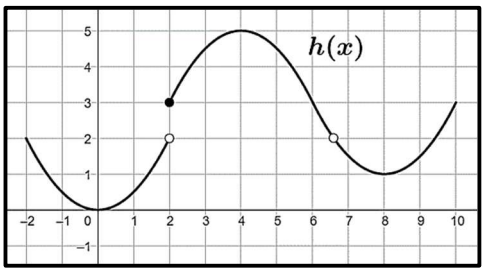
6. The graph of $h(x)$ is given at the right. Find all x values in the open interval $(-2, 10)$ where $h(x)$ is not differentiable.



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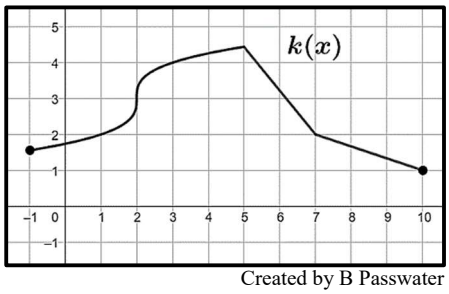
7. The graph of $k(x)$ is given at the right. Find all x values in the open interval $(-1, 10)$ where $k(x)$ is not differentiable.



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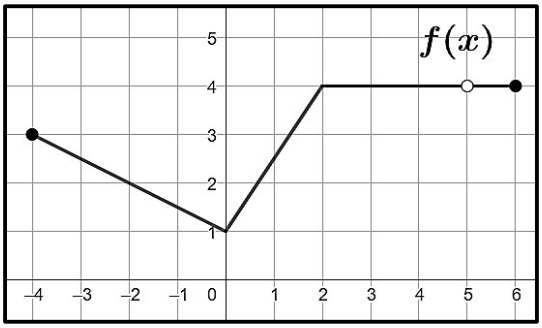
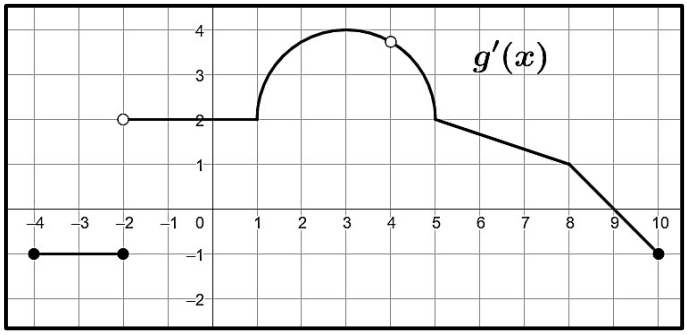


The function $f(x)$ is shown in the graph above. Find the following values, if possible.

8. $f'(-2)$ 9. $f'(0)$ 10. $f(2)$ 11. $f'(3)$

12. $f(0)$ 13. $f'(5)$ 14. $f'(-3)$ 15. $f'(1)$



The graph of $g'(x)$, the derivative of g, is given above. Find the following values, if possible.

16. $g'(5)$ 17. $g''(5)$ 18. $g''(-1)$ 19. $g''(-2)$

20. $g''(7)$ 21. $g'(9)$ 22. $g''(8)$ 23. $g'(-2)$

24. Put the following values in order from least to greatest from the graphs of f and g'

a. $f'(-1), f'(1), f'(3)$ b. $g'(3), g'(6), g'(2)$

c. $f(3), f(0), f'(-2.5)$ d. $g''(9), f'(-3), g'(2)$